Basic Info.

Model NO.	TCZY-2021
Warranty	1 Year
Detection Method	Spectrophotometer
Advantage	High Resolution
Principle	Classic Spectrometer
Wavelengths	UV-Vis-Nir Spectrometer
Dispersive Element	Grating Spectrometer
Light Number	Monochromator
Certification	ISO9001, CE
Customized	Customized
Wavelength Accuracy	±0.3 Nm
Wavelength Repeatability	≤0.1 Nm
Spectral Bandwidth	1, 2, 5, 10 and 50 Nm

Baseline Drift	0.0004 a/H
Dispersion Element	Grating
Installation Type	Laboratory Equipment
Transport Package	Export Wooden Case
Specification	Wavelength Range: 190-2800 nm;
Trademark	TANTODA
Origin	Hebei, China

Packaging & Delivery

Package Size	120.00cm * 163.00cm * 130.00cm
Package Gross Weight	100.000kg

Product Description

Application

The TCZY-2021 Building Glass Optical Performance Tester is designed for evaluating the optical and solar-energy properties of architectural glass. It is suitable for testing ordinary glass, coated glass, Low-E glass, solar-control glass and other energy-saving glazing products.

The instrument can be used to measure or calculate visible light transmittance and reflectance, UV transmittance, solar direct transmittance, solar absorptance, total solar energy transmittance, solar heat gain coefficient and shading coefficient.

It is widely applicable to architectural glass manufacturers, window and door companies, curtain wall manufacturers, construction material laboratories, quality inspection centers, universities, research institutes and third-party testing organizations. The instrument supports product development, production quality control, energy-efficiency evaluation and comparative analysis of different glass samples.

Product Description

The TCZY-2021 Building Glass Optical Performance Tester is a high-precision instrument designed to evaluate the optical and solar-energy properties of architectural glass. With a wavelength range of 190–2800 nm, it covers ultraviolet, visible and near-infrared regions.

The instrument measures spectral transmittance and reflectance and calculates parameters such as visible light transmittance, UV transmittance, solar direct transmittance, solar absorptance, total solar energy transmittance, solar heat gain coefficient and shading coefficient.

Featuring automatic wavelength calibration, selectable spectral bandwidth, low stray light and stable baseline performance, the TCZY-2021 is suitable for glass manufacturers, quality inspection laboratories, research institutes and building energy-efficiency testing organizations.

Key Features

Wide 190–2800 nm wavelength range covering UV, visible and near-infrared regions.
Measures transmittance and reflectance of architectural and energy-saving glass.
Calculates shading coefficient, solar heat gain coefficient and total solar energy transmittance.
Automatic wavelength calibration at startup for convenient daily operation.
High wavelength accuracy of ±0.3 nm and repeatability of ≤0.1 nm.
Selectable spectral bandwidths of 1, 2, 5, 10 and 50 nm.
Low stray light of less than 0.01% for reliable optical measurements.
Stable baseline performance with low drift and low noise.
Supports specular reflectance testing with the appropriate optical accessory.
Suitable for coated glass, Low-E glass, solar-control glass and other architectural glazing products.

Specification

Product Model	TCZY-2021
Measurement Method	Optical Measurement
Wavelength Range	190–2800 nm
Wavelength Accuracy	±0.3 nm
Wavelength Calibration	Automatic Calibration at Startup
Wavelength Repeatability	≤0.1 nm

Spectral Bandwidth	1, 2, 5, 10 and 50 nm
Stray Light	<0.01% at 220 nm (NaI) and 340 nm (NaNO2)
Photometric Range	−4.0 to 4.0 A
Photometric Accuracy	±0.002 A (0–0.5 A); ±0.004 A (0.5–1 A)
Photometric Repeatability	±0.3%T (0–100%T)
Absorbance Repeatability	0.001 A (0–0.5 A); 0.002 A (0.5–1 A)
Baseline Flatness	±0.001 A (190–1000 nm); ±0.005 A (1000–2800 nm)
Baseline Drift	0.0004 A/h at 500 nm and 0 A after warm-up
Noise Level	±0.0004 A
Optical Accessory	Specular Reflection Test Accessory
Packing Dimensions	990 × 680 × 480 mm